GROUP 4:

NLPify

Project Title:

Smart Sentiment: Analyzing Product Reviews for Business Insights

ITM 454 002 - Natural Language Processing
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Problem Statement
Online shopping platforms receive thousands of reviews every day. These reviews show what customers like or dislike, but most companies do not have the time or resources to read them all. As a result, important feedback may be ignored, and businesses cannot react quickly to improve their products or services. Manually analyzing reviews is slow and inefficient. Therefore, we need an automated system that can read and classify customer reviews into positive, negative, or neutral categories. This system will help businesses save time, understand customer opinions better, and make decisions based on real data.

Proposed Approach and Methodology
Feature Extraction: TF-IDF (Term Frequency – Inverse Document Frequency) is applied to convert reviews into numerical form so a model can process it. It reduces the impact of very common words (like "a," "the," or "and") and gives more weight to words that carry meaningful information for classification.
Model Selection: The primary model for this system is Multinomial Naive Bayes, which can assume words are independent and work efficiently with TF-IDF features for large datasets. To achieve the best performance, Logistic Regression is also evaluated for comparison, helping select the model with the best accuracy and balance across positive, negative, and neutral reviews.

Dataset Description
For this project, the dataset used is the Amazon Reviews for Sentiment Analysis dataset available on Kaggle (www.kaggle.com/datasets/bittlingmayer/amazonreviews). This dataset contains millions of customer reviews from Amazon across various product categories such as electronics, fashion, books, and home appliances. Each review includes both the written comment and a numerical star rating (from 1 to 5) which is mapped into three sentiment classes: positive (4–5 stars), neutral (3 stars), and negative (1–2 stars).

This mapping ensures a balanced classification system that captures favorable, unfavorable, and indifferent customer opinions. The wide range of product types also helps prevent the model from being biased toward a single product domain. Before training, the dataset will be preprocessed using:

Text Cleaning: removing HTML tags, punctuation, special characters, and unnecessary numbers.
Lowercasing: converting all words to lowercase for consistency.
Stopword Removal: filtering out common words (e.g., "the," "is," "and") that do not contribute to sentiment meaning.
Tokenization: splitting reviews into words or tokens for feature extraction.
Lemmatization: To normalize words to their base forms.

The data will be divided into training (80%) and testing (20%) sets to enable proper evaluation. An additional validation split may also be used during model development for tuning hyperparameters. This structure ensures that the trained models can be tested fairly on data they have not previously seen.

Expected Outcomes and Evaluation Metrics
Primary Goal: Expect the model to produce at least 80% accuracy for both Naive Bayes and Logistic Regression models, with balanced precision between positive, negative, and neutral classes.
Evaluation Metrics:

- **Accuracy:** overall percentages of correctly predicted reviews (expected >= 80%).
- **Precision:** minimize the false positive rate for fake positive and negative reviews prediction.
- **Recall:** maximize the true positive rate for identifying actual positive and negative reviews prediction.
- **F1-Score:** balanced measure considering both precision and recall.